

JOEAL BIJOY THOMAS

Data Scientist | Data & Business Analyst | Machine Learning & Generative AI

Newark, DE | joealthomas4@gmail.com | LinkedIn | Github | Portfolio

PROFESSIONAL SUMMARY

Data Scientist and Business Analyst who translates ambiguous product and business questions into actionable analyses and machine learning solutions. Skilled in SQL, Python, R, statistical testing, forecasting, cohort analysis, KPI development, and data visualization, with experience gathering requirements and presenting insights to cross-functional stakeholders.

TECHNICAL SKILLS

AI & Generative AI: Generative AI & LLM Integration, Deep Learning & Neural Networks (PyTorch, TensorFlow, CNNs, Vision Transformers), Machine Learning & Predictive Modeling, Data Engineering & ETL Pipelines for AI, Applied AI Model Deployment

Programming & Databases: Python (Pandas, NumPy, scikit-learn, PyTorch, Matplotlib, PyQt), R, SQL (Joins, CTEs, Window Functions), JavaScript, NoSQL, SAS, Stata, MATLAB

Machine Learning & Statistics: Regression, Classification, Time-Series Forecasting, Churn & Cohort Analysis, A/B & Hypothesis Testing, Feature Engineering, Deep Learning, NLP

Data Engineering & Analysis: ETL Pipelines, Data Cleaning & Wrangling, EDA, Dimensional/Star-Schema Modelling, N8N, Git

BI & Visualization: Power BI, Tableau, DAX, Power Query, Advanced Excel, Google Sheets

Business & Product: KPI Definition & Tracking, Requirements Gathering, Stakeholder Management, Executive Reporting, Data Storytelling, Program Evaluation, Product Analytics

PROFESSIONAL EXPERIENCE

University of Delaware, Audio Video Operations & Event Technology Technician

Jan 2026 – Present

- Analyze operational data across 20+ technicians and 50+ annual events, defining and tracking KPIs such as utilization, turnaround time, and incident rate to support staffing decisions and ongoing program evaluation.
- Manage stakeholder communication and requirements gathering across event, academic, and facilities teams to align operational needs with data-informed decisions.

The Blue Fox Media, Junior Data Analyst, Pune, India

Dec 2024 – Jul 2025

- Wrote complex SQL (joins, CTEs, window functions) to extract and transform large datasets, improving query performance by 40%.
- Ran EDA in Python using Pandas and NumPy and built Power BI dashboards with data storytelling for leadership, cutting reporting turnaround by 25% and contributing to a 20% increase in ROI.

Dentsu, Associate Business Analyst (Contract), Baner, India

May 2024 – Nov 2024

- Managed a \$75,000 monthly budget across Google Ads and AppsFlyer campaigns, applying product analytics and predictive audience models while gathering cross-functional requirements to drive 3–4× subscription growth.
- Optimized campaigns against Target Cost Per Acquisition (TCPA) as the primary KPI, designing cohort-level A/B tests that reduced CAC by 30–40% and delivered 2–3× revenue growth through data-driven strategies.
- Automated reporting workflows in JavaScript and Excel, cutting manual effort by 80% and improving targeting precision by 75%.

Indira College of Engineering & Management, Research Software Engineer – Data Science (Focused), Pune, India

Sep 2022 – Apr 2024

- Processed and engineered features across multiple large-scale datasets for vehicle and motion detection, feeding downstream computer vision and ML models for object classification and movement tracking.
- Built and shipped a desktop Indian Food Image Recognition application in PyQt and Python3, integrating a fine-tuned Vision Transformer (ViT) model achieving 94% classification accuracy.
- Designed and maintained a NoSQL data layer to support real-time recipe and calorie-lookup queries within the application.

PROJECTS

Delaware Predictive Hourly Insights (DELPHI), Hackathon Project

GitHub – Project Link

- Engineered an end-to-end quantitative forecasting pipeline using LightGBM and a from-scratch NumPy Graph Neural Network (GNN), producing a validated 170,956-row submission across 118 stations, including 43 cold-start stations.
- Diagnosed critical data leakage and applied split-conformal calibration for statistical validation, achieving 90.0% coverage and GEH < 10 at 54% accuracy.
- Deployed the predictive pipeline with an interactive AI chatbot on AWS and Amazon Bedrock, translating forecasts into real-time route optimization recommendations for transportation stakeholders.

Autoregressive Normalizing Flows for Density Estimation & Anomaly Detection

GitHub – Project Link

- Implemented RealNVP and MAF generative models from scratch in PyTorch (Zuko), verifying invertibility to machine precision and evaluating log-likelihoods as anomaly-detection scores on 64-D digit images.

Credit Risk Intelligence Dashboard

GitHub – Project Link

- Designed a star-schema model for 30,000 credit-card customers across 25 variables and built default-rate KPIs by credit tier using DAX (AVERAGE, ALLEXCEPT), surfacing 50+ high-risk outlier customers.

EDUCATION

University of Delaware, M.S., Data Science, Newark, DE

2027

Coursework: Regression Analysis, Analysis and Design of Algorithms, Applied Multivariate Statistics, Database Systems, Big Data Technologies, Mathematical Techniques in Data Science

Indira College of Engineering and Management, B.E., Computer Engineering, Pune, India

2024